

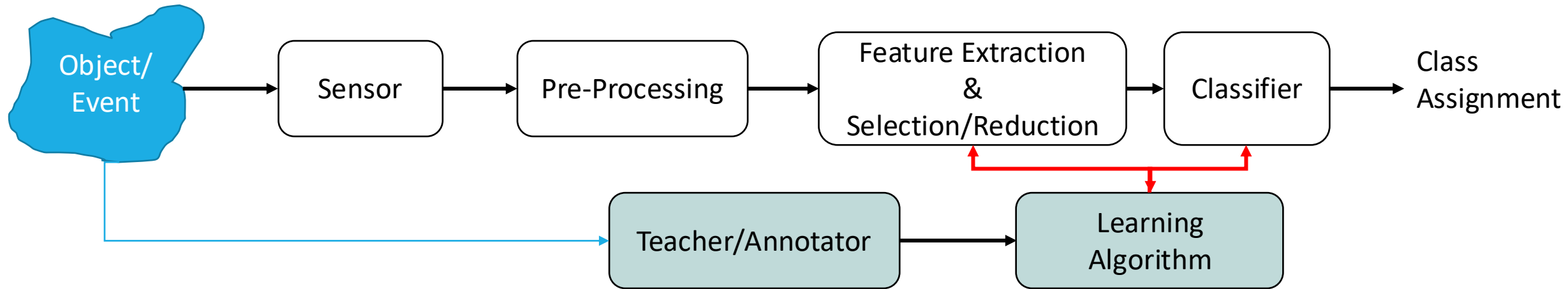
Aprendizagem Computacional (AC) Machine Learning (ML)



Introduction (II)

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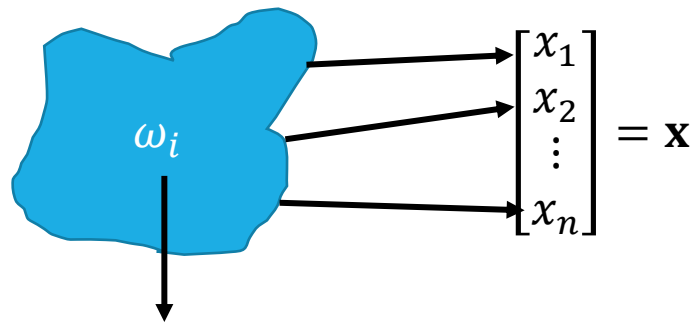
Components of a PR System



- **Teacher/Annotator:** Provides information about the relation between the objects and related labels used for supervised learning/training
- **Learning Algorithm:** Sets the classifier and feature extraction parameters based on the training data

Representation

Object/Event

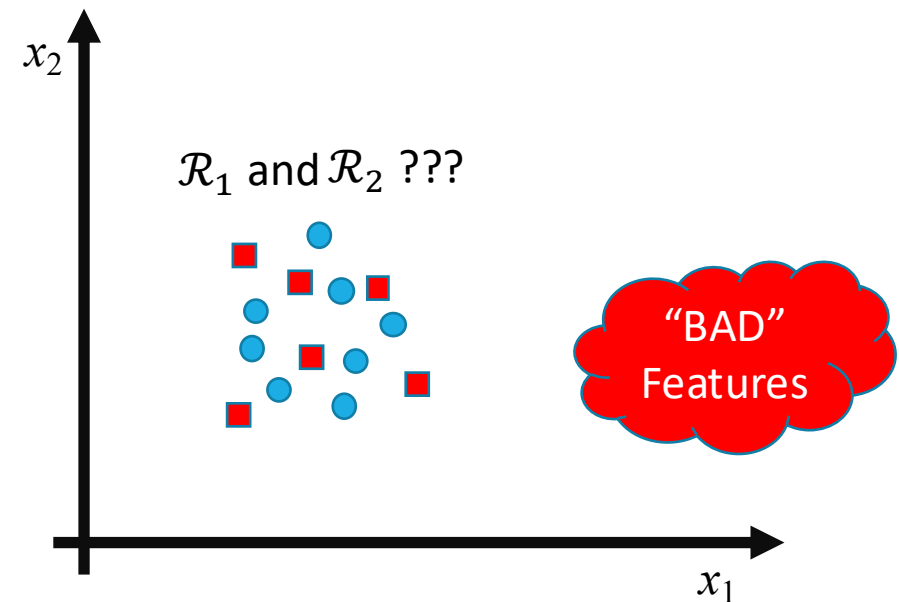
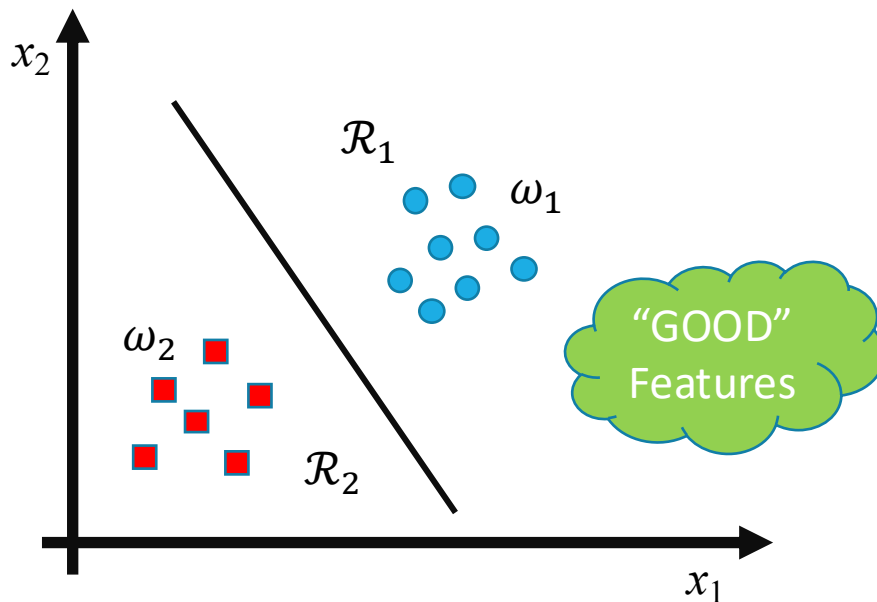


- Feature vector $\mathbf{x} \in X$
- $\mathbf{x}$ is a point on the **feature (representation) space** X .

- Object class $\omega_i \in \Omega$ ($i = 1, \dots, K$), where Ω is the set of all c classes that defines the **interpretation space**.
- *Classification Task*:
 - Design a decision function $d(\mathbf{x}): X \rightarrow \Omega \rightarrow$ **Mapping between the feature space and the interpretation space**
 - Decides about the class of a given object.

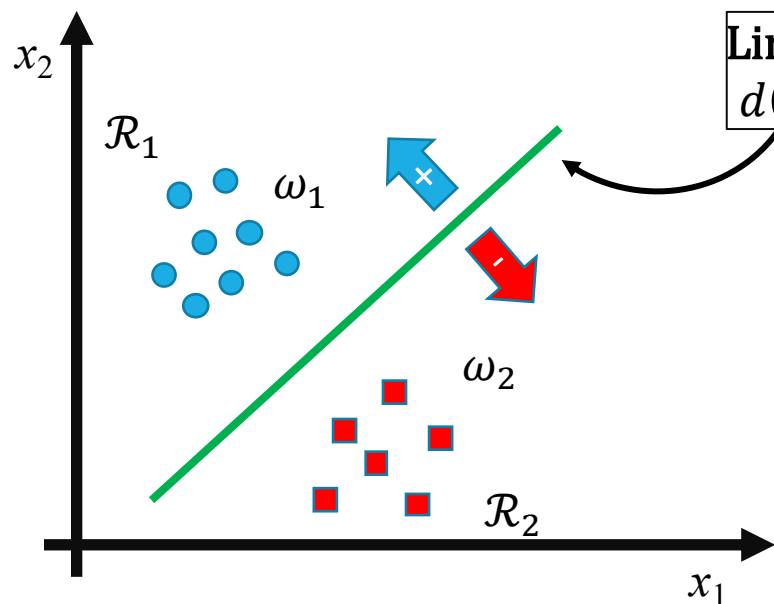
Representation

- Features should be good for classification.
- Good features:
 - Objects from the same class have similar feature values.
 - Objects from different classes have different values.



Linear Decision Functions

- Consider a set of objects/events represented by feature vectors.
- For simplicity we assume a representation in a $\mathbb{R}^2$ where each object/event is represented by:
 - $\mathbf{x} = [x_1 \ x_2]^T$
- Each object can be separated in one of two classes ω_1 and ω_2 .
- To classify we need to split the representation space in decision regions $\mathcal{R}_i$. $\rightarrow$ Decision function $d(\mathbf{x})$.



Linear Decision Function:

$$d(\mathbf{x}) = w_1x_1 + w_2x_2 + w_0 = \mathbf{w}^T \mathbf{x} + w_0 = \mathbf{w}^T \mathbf{x} - b$$

- $\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \rightarrow$ Vector Normal to the decision (hyper)plane
- $w_0 = -b$

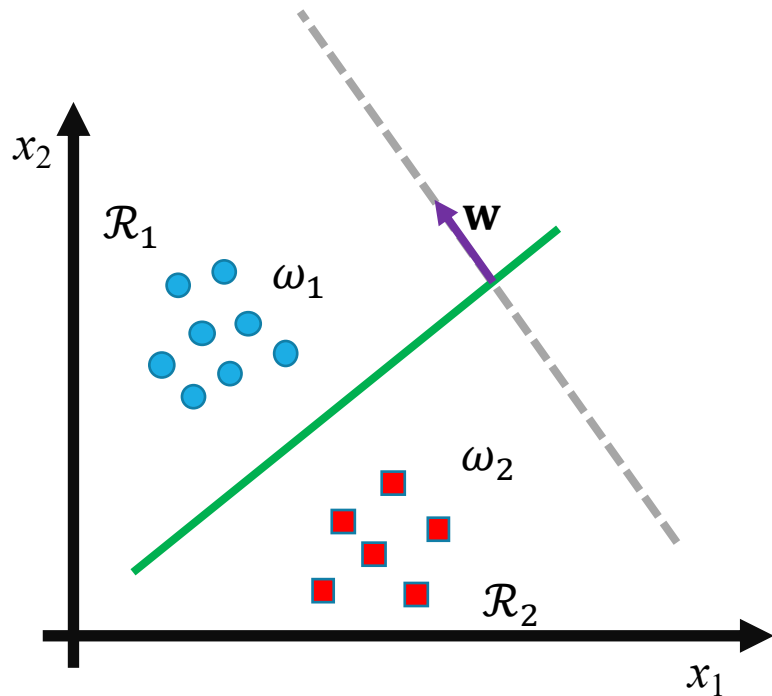
Linear Classifier:

$$decision = \begin{cases} \omega_1 & \text{if } d(\mathbf{x}) \geq 0 \\ \omega_2 & \text{if } d(\mathbf{x}) < 0 \end{cases}$$

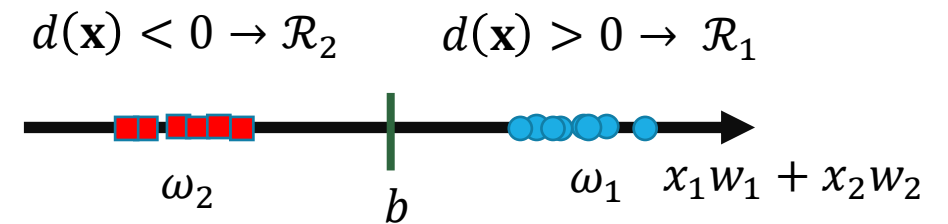
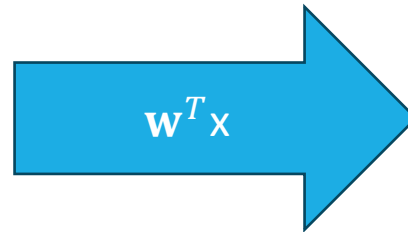
Linear Decision Functions

Linear Decision Function:

$$d(\mathbf{x}) = \mathbf{w}^T \mathbf{x} - b$$



$\mathbb{R}^2$ ($\mathbb{R}^D$)



$\mathbb{R}^1$

Classification: Generalized Linear Decision Function

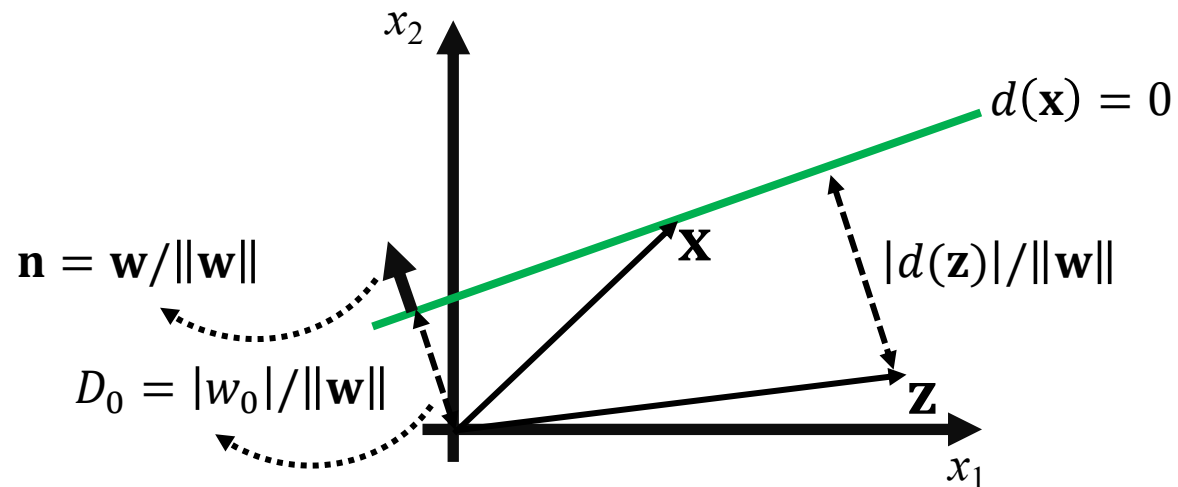
• In $\mathbb{R}^d$: $d(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + \dots + w_Dx_D = \mathbf{w}^T \mathbf{x} + w_0 = \mathbf{w}^{*T} \mathbf{x}^*$

where:

❖ $\mathbf{w} = [w_1 \quad \dots \quad w_D]^T \rightarrow$ Weight Vector

❖ $\mathbf{w}^* = [w_0 \quad w_1 \quad \dots \quad w_D]^T \rightarrow$ Augmented Weight Vector, including bias

❖ $\mathbf{x}^* = [1 \quad x_1 \quad \dots \quad x_D]^T \rightarrow$ Augmented Feature Vector



Distance from a point to a plane:

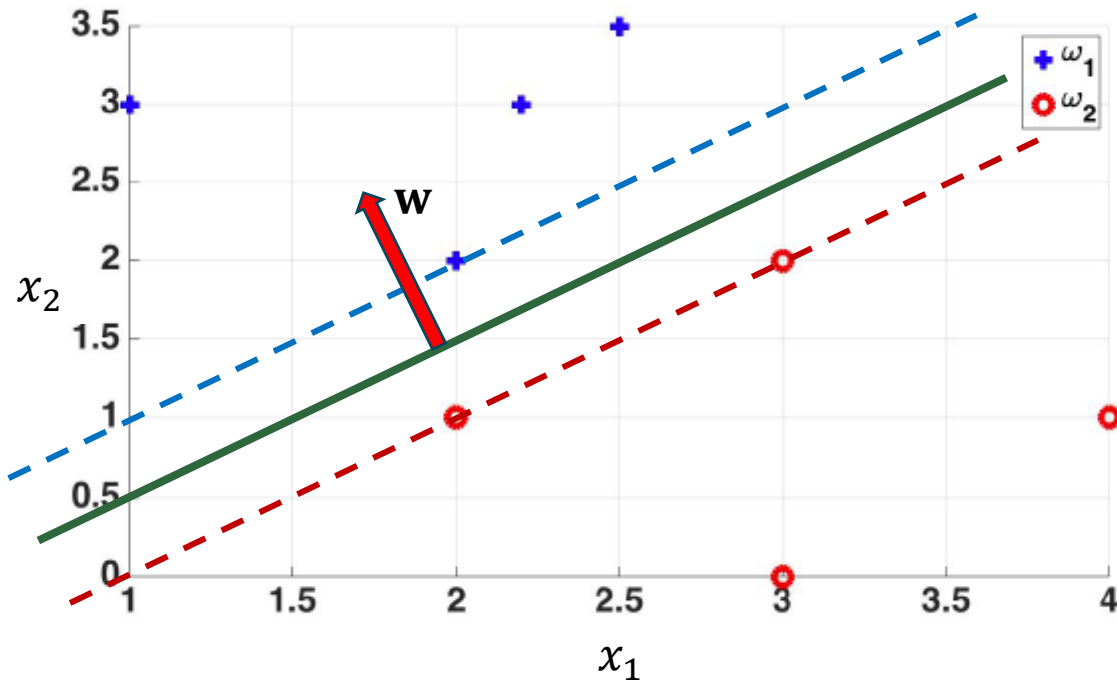
$$D = \frac{|w_0 + w_1x_1 + w_2x_2 + \dots + w_Dx_D|}{\sqrt{w_1^2 + w_2^2 + \dots + w_D^2}} = \frac{|d(\mathbf{x})|}{\|\mathbf{w}\|}$$

- D_0 (Distance to the origin) $\rightarrow$ Define the deviation to the origin
- $\mathbf{n}$ (normal vector in $d(\mathbf{x}) > 0$ direction) $\rightarrow$ Define the slope

Exercise

- Find a linear decision hyperplane, such that $d(\mathbf{x}) > 0$ for ω_1 ?

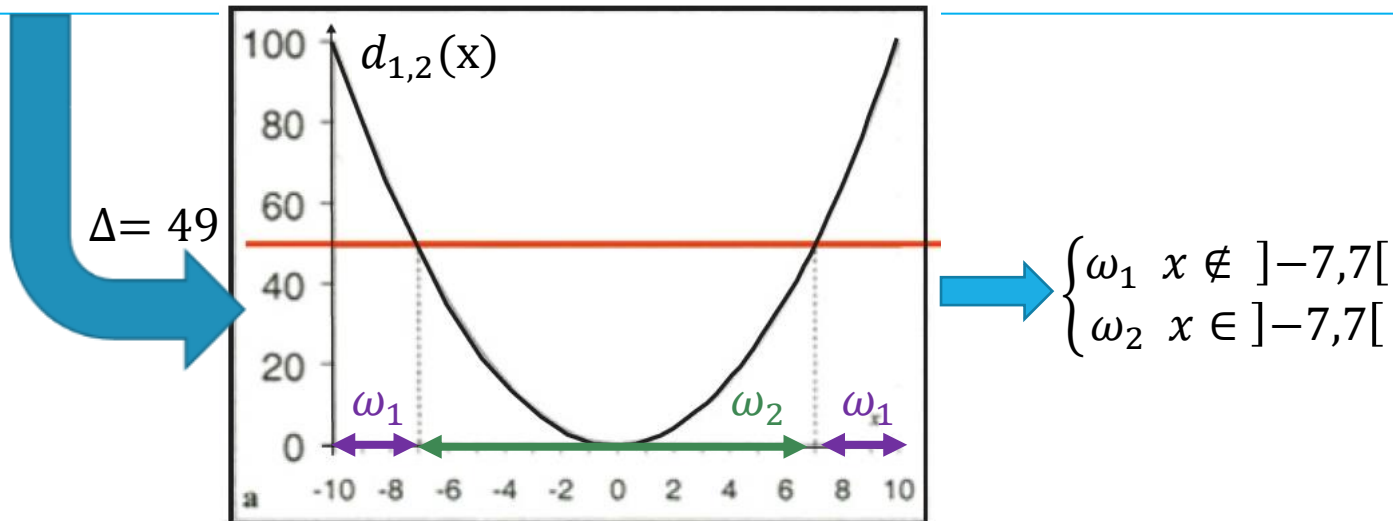
$$d(\mathbf{x}) = w_1 x_1 + w_2 x_2 - w_0$$



- Magenta line $\rightarrow x_2 = m_m x_1 + b_m$
 - $m_m = \frac{2-1}{3-2} = 1$
 - Use one of the points to find b_m . Lets use (2,1) $\rightarrow 1 = 2 + b_m \Leftrightarrow b_m = -1$
 - $x_2 = x_1 - 1$
- Blue line $\rightarrow x_2 = m_b x_1 + b_b$
 - Slope is the same $m_b = m_m = 1$
 - Use point (2,2) to find $b_b \rightarrow 2 = 2 + b_b \Leftrightarrow b_b = 0$
 - $x_2 = x_1$
- Lets assume that a good hyperplane lies in the middle of the two lines
 - Same slope and diferente $b \rightarrow$
 $x_2 = x_1 + \frac{b_m + b_b}{2} \Leftrightarrow x_2 = x_1 - 0.5$
 - Decision: $d(\mathbf{x}) = -x_1 + x_2 + 0.5 = 0$
 - $d(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0$ with $\mathbf{w} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, w_0 = 0.5$

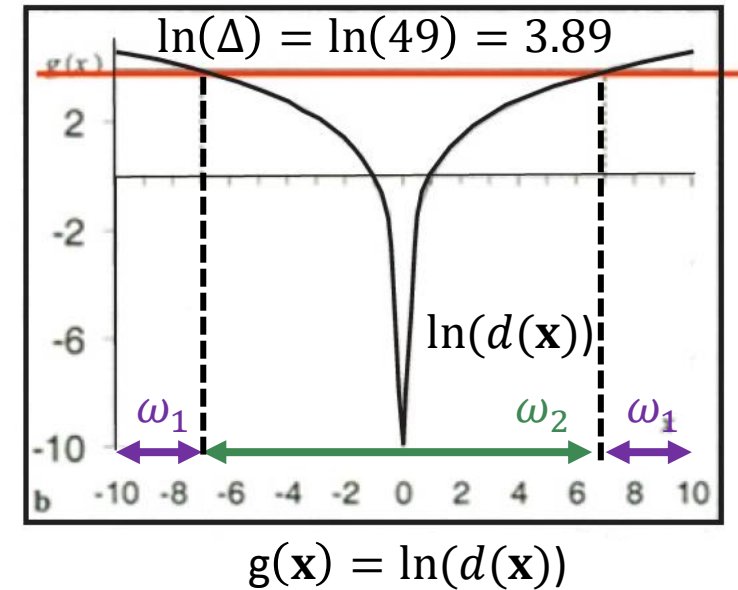
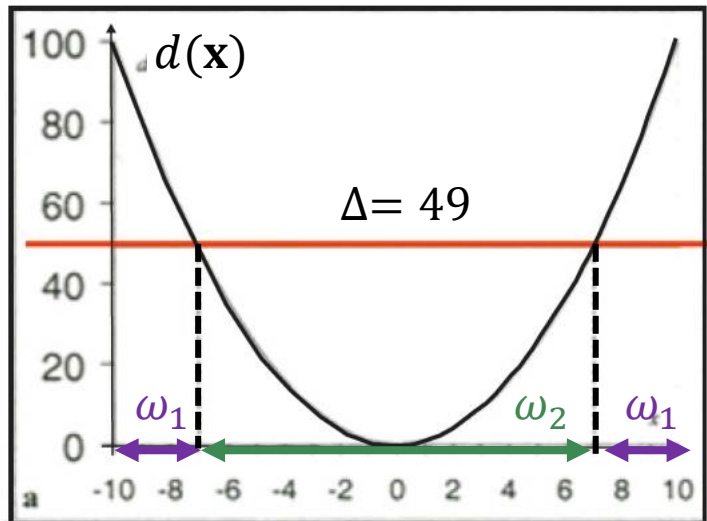
Classification: Generalized Decision Function

- We are **not limited to linear decision** hyperplanes.
 - ❖ $d_{i,j}(\mathbf{x}) = f(\mathbf{x}) > 0$ if $\mathbf{x} \in \omega_i$; $d_{i,j}(\mathbf{x}) = f(\mathbf{x}) < 0$ if $\mathbf{x} \in \omega_j$ with $j \neq i$
- Thresholds, Δ , different from 0 can also be used:
 - ❖ $d_{i,j}(\mathbf{x}) > \Delta$ if $\mathbf{x} \in \omega_i$; $d_{i,j}(\mathbf{x}) < \Delta$ if $\mathbf{x} \in \omega_j$ with $j \neq i$
- Example:
 - ❖ One feature, two class problem with a quadratic decision function $d(x) = x^2$:
if $d(x) = f(x) = x^2 \geq 49$ (Δ) then $x \in \omega_1$ else $x \in \omega_2$



Classification: Generalized Decision Function

- Any functional transformation of $d(\mathbf{x})$ by a **monotonic** function leads exactly to the same classification result. Example:
 - ❖ if $d(x) = x^2 \geq \Delta$ then $x \in \omega_1$ else $x \in \omega_2$ is equivalent to:
 - ❖ if $\ln(d(x)) = \ln(x^2) \geq \ln(\Delta)$ then $x \in \omega_1$ else $x \in \omega_2$



Classification: Generalized Decision Function

- Generalized decision functions as a functional linear combination:

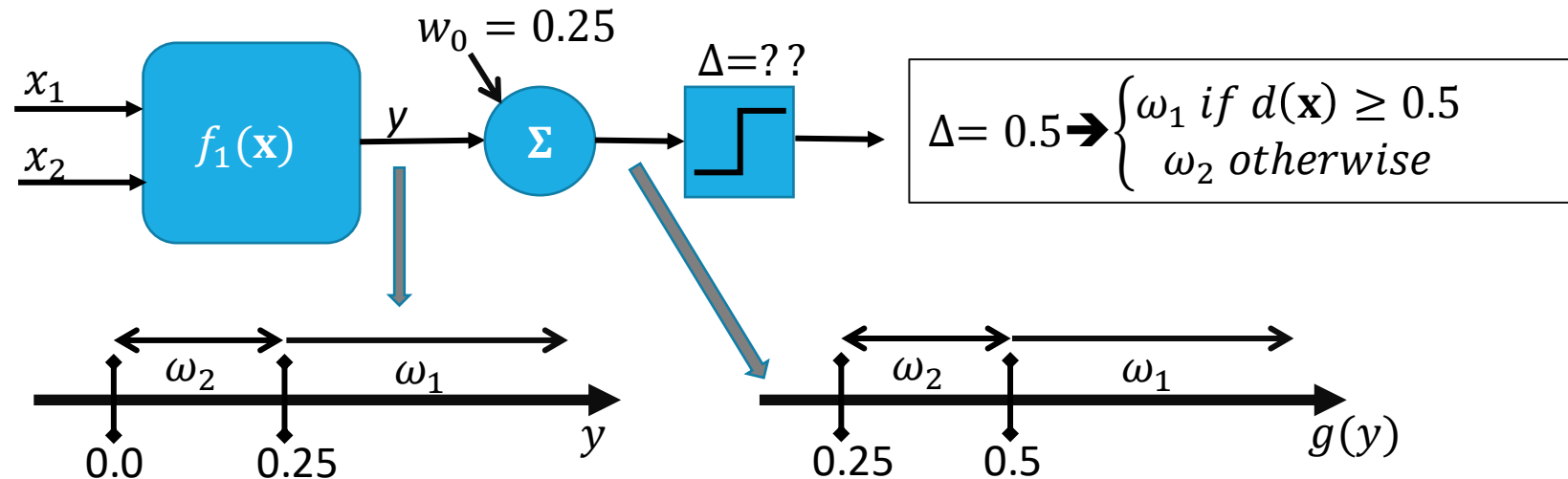
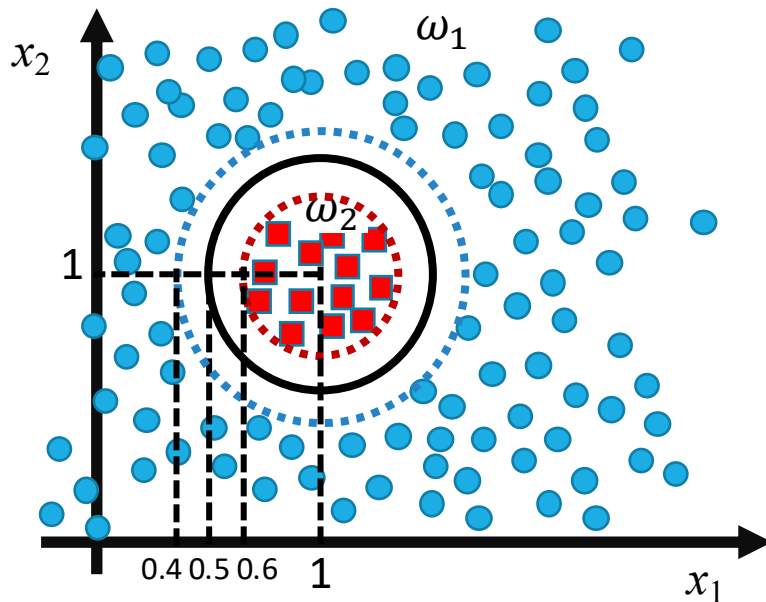
❖ $d(\mathbf{x}) = w_1 f_1(\mathbf{x}) + w_2 f_2(\mathbf{x}) + \cdots + w_k f_{D'}(\mathbf{x}) + w_0 = \mathbf{w}^T \mathbf{y} + w_0$ with $\mathbf{y}(\mathbf{x}) = [f_1(\mathbf{x}) \ f_2(\mathbf{x}) \ \cdots \ f_{D'}(\mathbf{x})]^T$ and $\mathbf{w} = [w_0 \ w_1 \ \cdots \ w_{D'}]^T$

Decision function in the **original space**: $d(\mathbf{x}) = (x_1 - 1)^2 + (x_2 - 1)^2 + 0.25$

- Example:

$d(\mathbf{x}) = (x_1 - 1)^2 + (x_2 - 1)^2 + 0.25 = \mathbf{w}^T \mathbf{y}(\mathbf{x}) + w_0$
with $\mathbf{w} = [1]^T$, $w_0 = 0.25$, and $\mathbf{y}(\mathbf{x}) = [f_1(\mathbf{x})]^T$ where $f_1(\mathbf{x}) = (x_1 - 1)^2 + (x_2 - 1)^2$

We can formulate the decision in a one-dimensional **transformed space**:
 $g(y) = y + 0.25$ (**Linear decision function**)



Classification: High Degree Polynomial Decision Function

- For a problem with two features ($D=2$) and a maximum polynomial of degree 4 ($O=4$), i.e., space of functions $x_1^p x_2^q$ with $p, q = [0,1,2,3,4]$:

$$\diamond d(\mathbf{x}) = w_0 + w_1 x_2 + w_2 x_1 + w_3 x_1 x_2 + w_4 x_2^2 + w_5 x_1^2 + w_6 x_1 x_2^2 + w_7 x_1^2 x_2 + w_8 x_2^3 + w_9 x_1^3 + w_{10} x_1^2 x_2^2 + w_{11} x_1 x_2^3 + w_{12} x_1^3 x_2^2 + w_{13} x_1^4 + w_{14} x_2^4$$

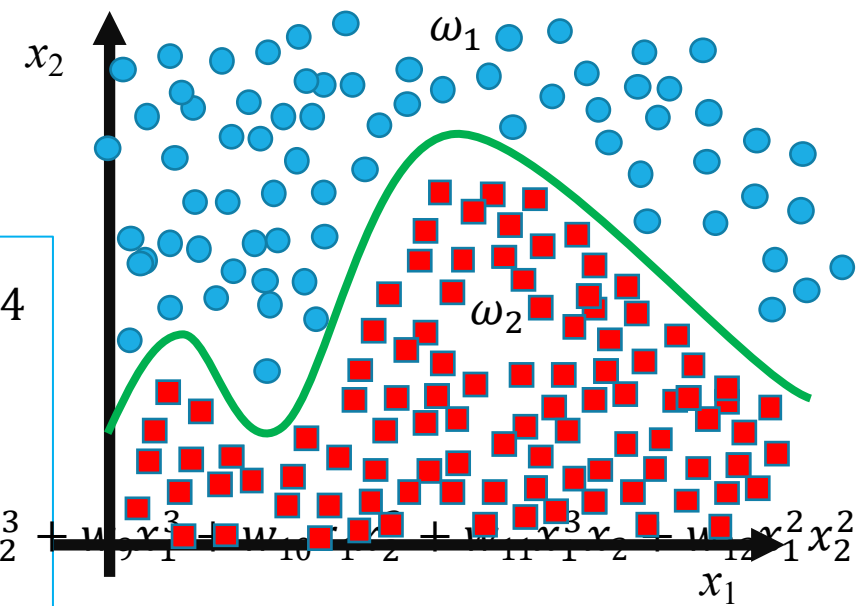
$w_0 =$

$$\mathbf{w}^T \mathbf{f}(\mathbf{x}) + w_0$$

❖ 15 weights have to be defined → 14 dimensions

- For a problem with dimensionality D and a polynomial with degree O , the number of free parameters are:

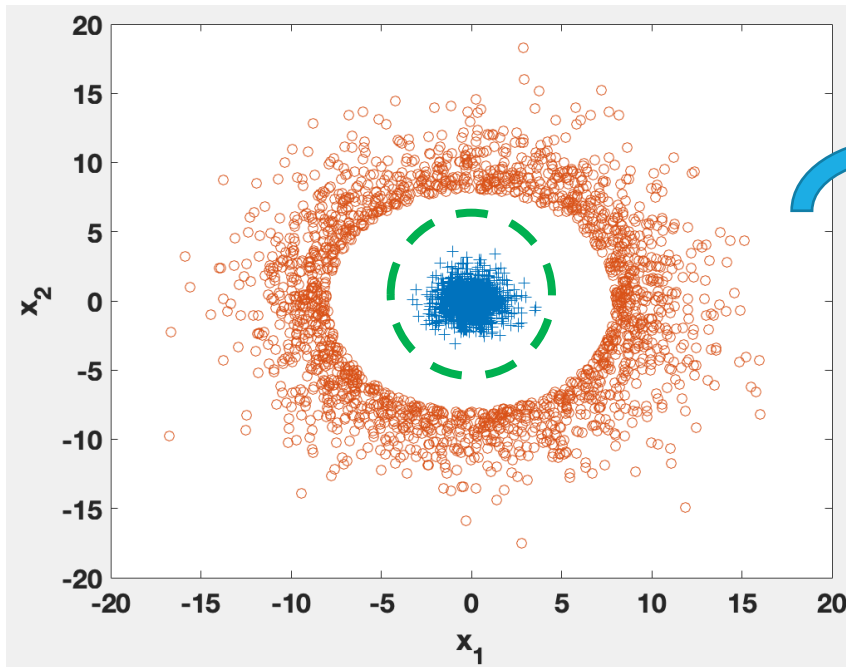
$$\diamond D' = C_O^{O+D} = \frac{(O+D)!}{O!D!} \rightarrow \text{May be a Huge Number}$$



- Representation in a high dimensional transformed space
 - Advantages: It might be possible to determine a linear decision function in the transformed space that separates data
 - Disadvantages: High dimensional spaces raises a number of difficulties (we will see later)

Classification: High Degree Polynomial Decision Function (Example)

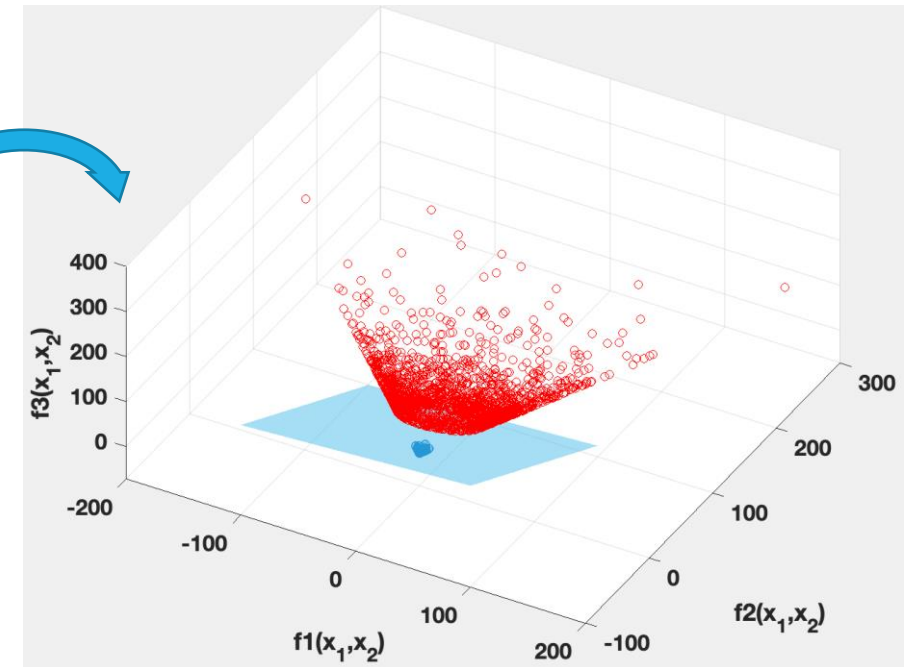
- Original Feature Space



Non-linearly separable ☹️

- High dimensional transformed space

$$f = \begin{bmatrix} f_1(x_1, x_2) \\ f_2(x_1, x_2) \\ f_3(x_1, x_2) \end{bmatrix} = \begin{bmatrix} x_1 x_2 \\ x_1^2 \\ x_2^2 \end{bmatrix}$$

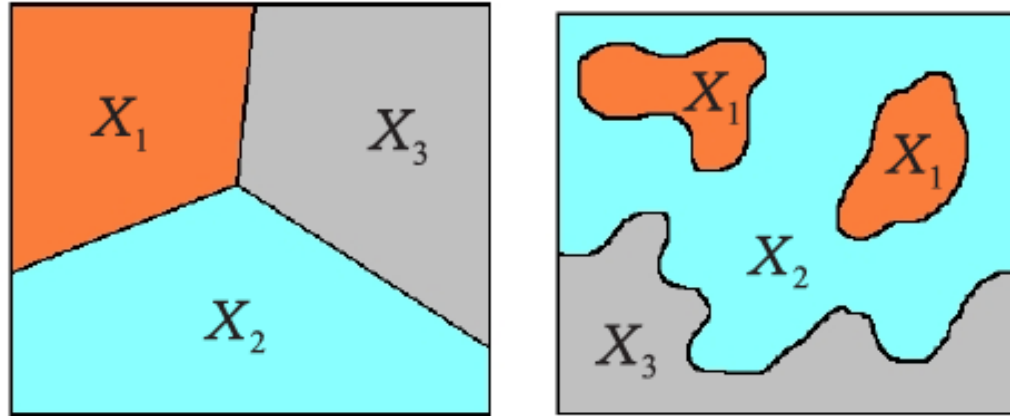


Linearly separable 😊

$$d\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = w^T f\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) + w_0 = \begin{bmatrix} w_1 & w_2 & w_3 \end{bmatrix} \begin{bmatrix} x_1 x_2 \\ x_1^2 \\ x_2^2 \end{bmatrix} + w_0$$

Classification: Multiclass

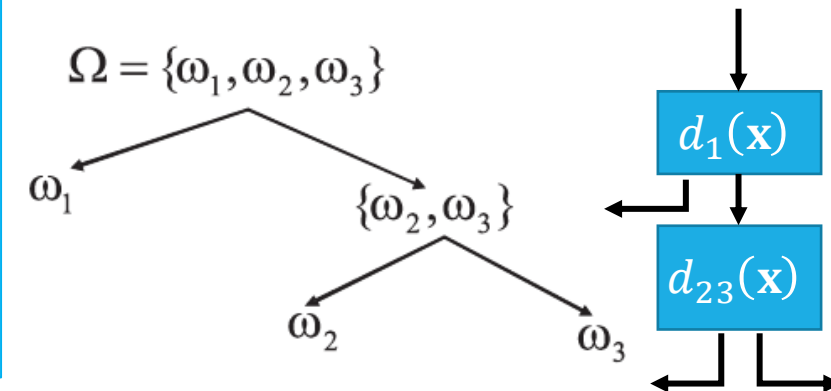
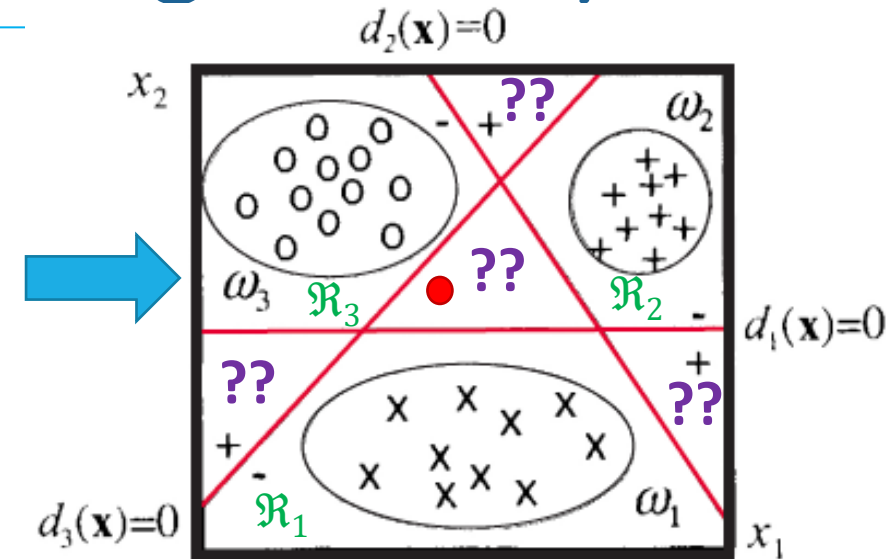
- Multiclass based on the combination of several binary decision functions



- Two paradigms exist that result in the consideration of several binary problems:
 - **Absolute separation** (One-Against-All or One-Versus-All)
 - **Pairwise separation** (One-Against-One or One-Versus-One)
- **Other approaches exist** that deal with multi-class problems without the need to consider several binary problems, e.g. **naive-bayes** (We will see latter)

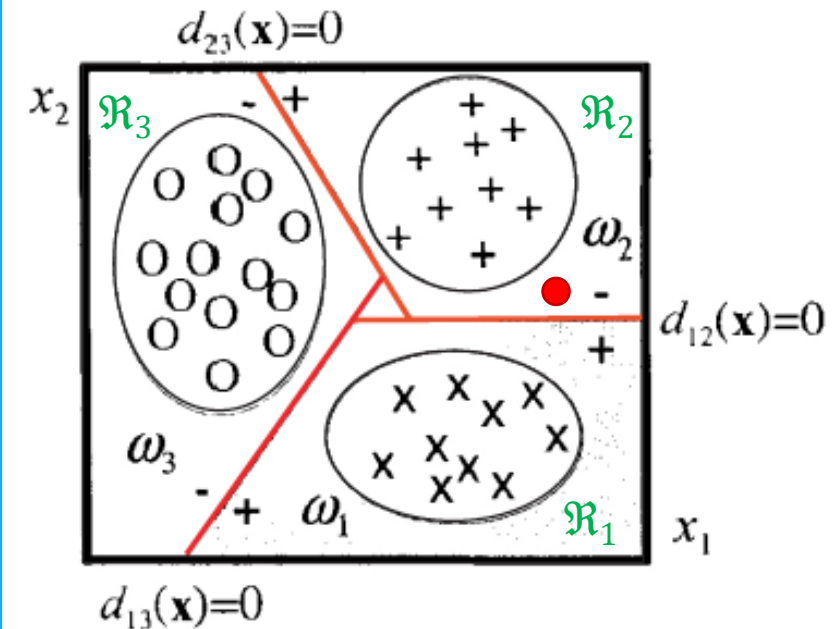
Classification: Multi-Class (One-Against-All)

- Considering as example (without loss of generality) a three-class problem ($K=3$) defined in a two-dimensional feature space ($D=2$), and considering linear decision surfaces
- OAA implies the development of K classifiers
- The decision regions are defined by:
 - ❖ $\mathcal{R}_i = \{\mathbf{x}; d_i(\mathbf{x}) > 0, d_j(\mathbf{x}) < 0, i, j = 1, \dots, K, j \neq i\}$
- Can be viewed as a hierarchical classification
→ **Solve the problem of inconclusive regions** → At each level a decision is made
 - $d_{23}(\mathbf{x}) = d_2(\mathbf{x}) - d_3(\mathbf{x})$



Classification: Multi-Class (One-Against-One)

- Classes may be separated pairwise
- $\binom{K}{2}$ decision functions separating pairs of classes:
 - ❖ $d_{ij}(\mathbf{x}) > 0 \quad \forall x \in \omega_i$ and $d_{ij}(\mathbf{x}) < 0 \quad \forall x \in \omega_j$ ($d_{ij}(\mathbf{x}) = -d_{ji}(\mathbf{x})$)
- The decision regions are defined by:
 - ❖ $\mathcal{R}_i = \{\mathbf{x}; d_{ij}(\mathbf{x}) > 0, \quad i, j = 1, \dots, K, \quad j \neq i\}$
- Final decision usually performed by a **maximum voting** approach

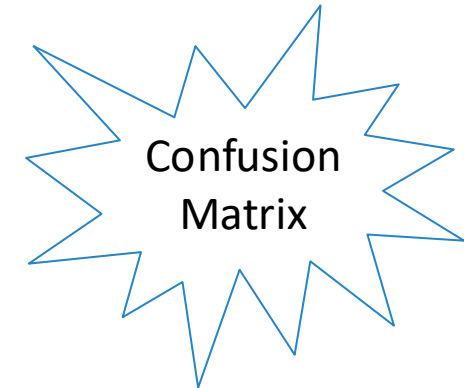


Performance assessment: Binary & Supervised

- Performance Assessment → Compare predicted with real class labels

$\omega_1 \rightarrow$ Positive
 $\omega_2 \rightarrow$ Negative

Observations		Predicted	
		Class ω_1	Class ω_2
True	Class ω_1	41 (TP)	9 (FN)
	Class ω_2	14 (FP)	36 (TN)



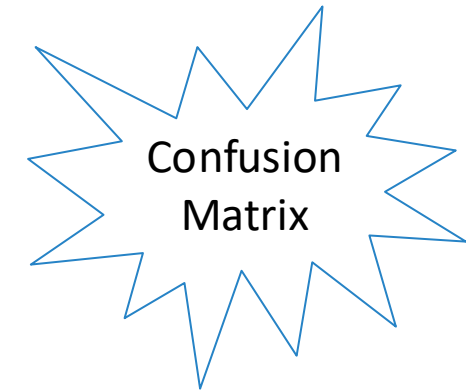
- Accuracy (%) = $\frac{TP+TN}{TP+TN+FP+FN} \times 100 = 77\%$
- Sensitivity (%) = $\frac{TP}{TP+FN} \times 100 = CC_{\omega_1}(\%) = 82\%$
- Specificity (%) = $\frac{TN}{TN+FP} \times 100 = CC_{\omega_2}(\%) = 72\%$
- Error (%) = $100 - \text{Accuracy} = 23\%$
- ω_1 error (%) = False Negative Rate (%) = $100 - \text{Sensitivity} = 100\% - 82\% = 18\%$
- ω_2 error (%) = False Positive Rate (%) = $100 - \text{Specificity} = 100\% - 72\% = 28\%$

Performance assessment: Binary & Supervised

- Performance Assessment → Compare predicted with real class labels

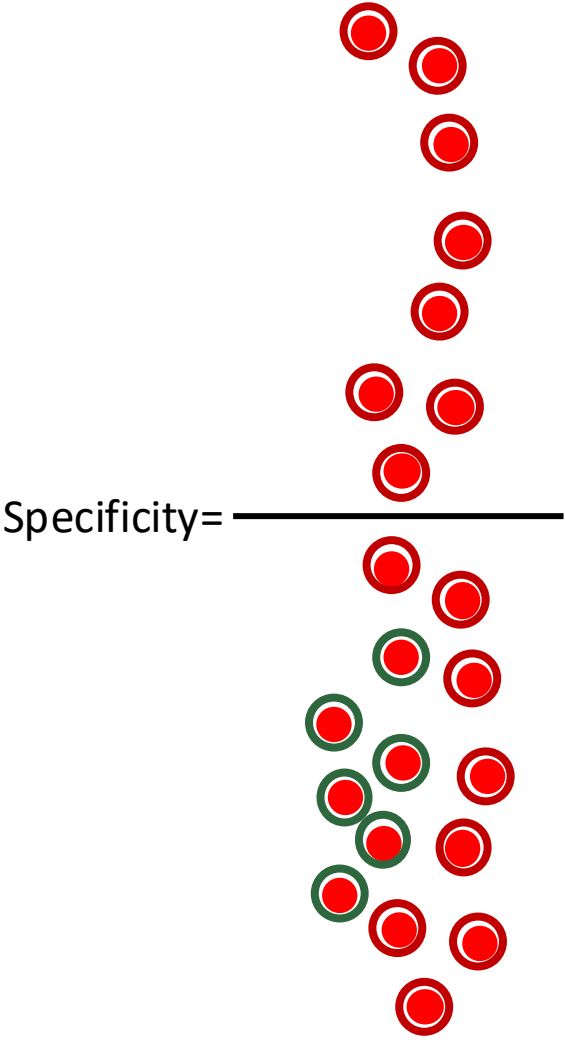
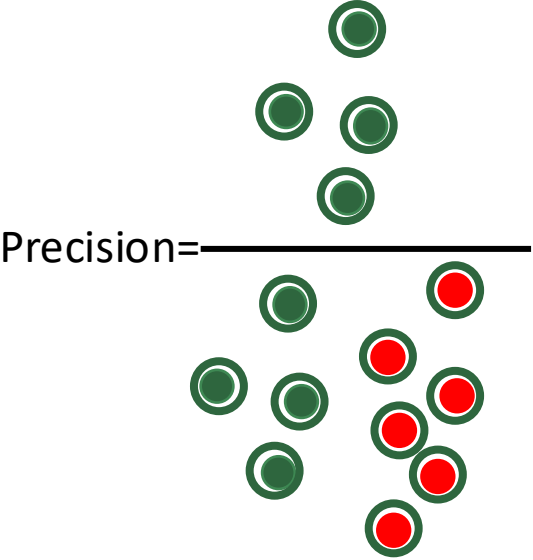
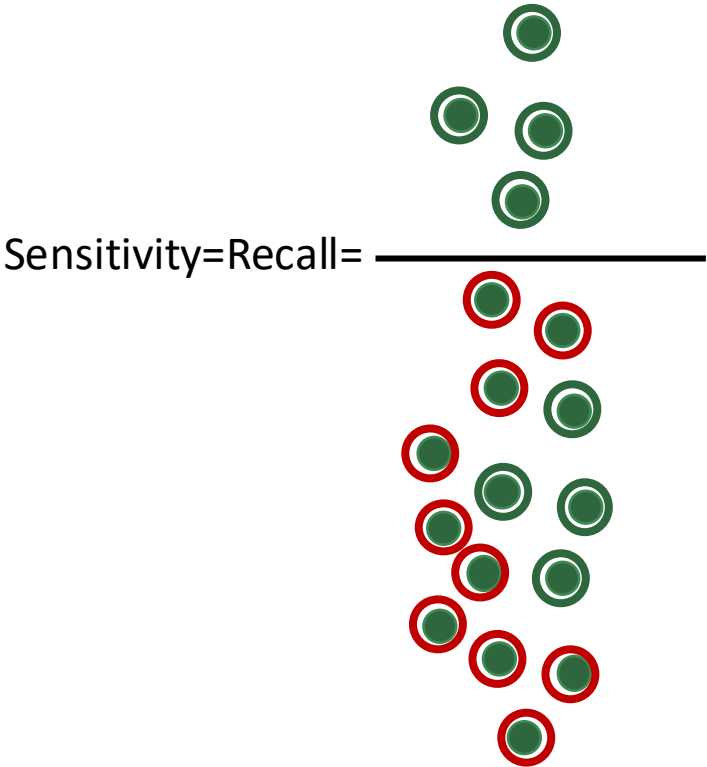
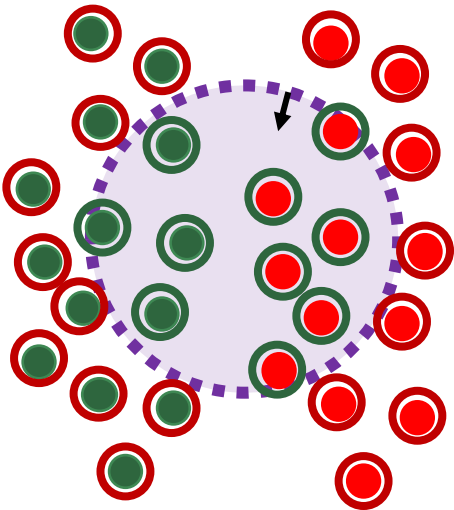
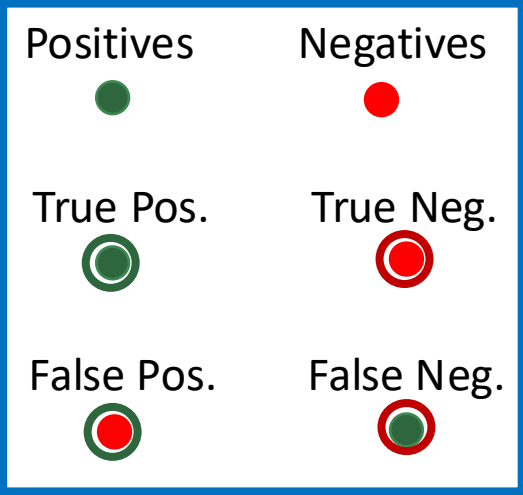
$\omega_1 \rightarrow$ Positive
 $\omega_2 \rightarrow$ Negative

Observations		Predicted	
		Class ω_1	Class ω_2
True	Class ω_1	41 (TP)	9 (FN)
	Class ω_2	14 (FP)	36 (TN)



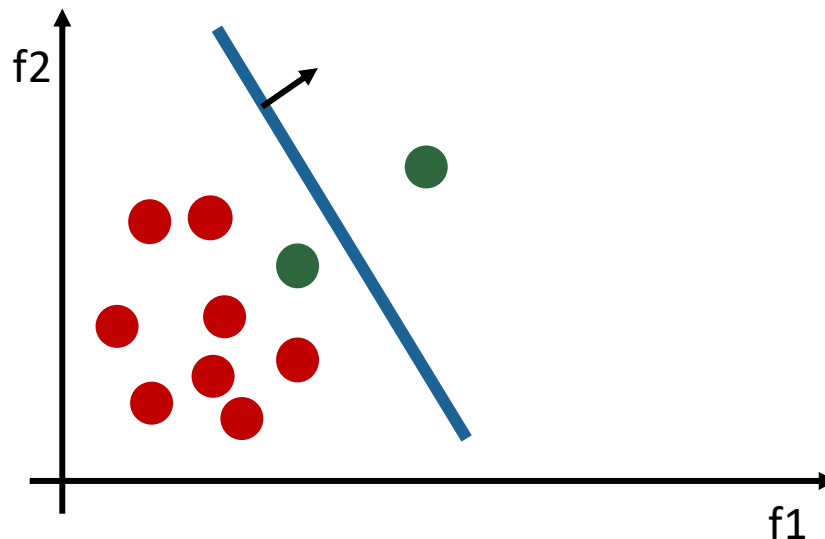
- $\text{Precision}(\%) = \frac{TP}{TP+FP} \times 100 = 74\%$
- $\text{Recall}(\%) = \text{Sensitivity}(\%) = \frac{TP}{TP+FN} \times 100 = \text{CC}_{\omega_1}(\%) = 82\%$
- $F1\text{score}(\%) = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 78\%$

Sensitivity, Specificity & Precision



Performance assessment: Binary Example

$\omega_1 \rightarrow$ Positive
 $\omega_2 \rightarrow$ Negative



TP=1
TN=8
FP=0
FN=1

- Accuracy (%) = ACC (%) = $\frac{TP+TN}{TP+TN+FP+FN} \times 100 = \frac{1+8}{1+8+0+1} \times 100 = 90\%$
- Sensitivity (%) = SS (%) = $\frac{TP}{TP+FN} \times 100 = \frac{1}{1+1} \times 100 = CC_{\omega_1}(\%) = 50\%$
- Specificity (%) = SP (%) = $\frac{TN}{TN+FP} \times 100 = \frac{8}{8+0} \times 100 = CC_{\omega_2}(\%) = 100\%$
- Precision (%) = PR(%) = $\frac{TP}{TP+FP} \times 100 = \frac{1}{1+0} \times 100 = 100\%$
- Recall(%) = Sensitivity (%) = SS(%) = 50%
- $F1score(\%) = 2 \frac{0.5 \times 1}{0.5 + 1} \times 100 \approx 66.7\%$

		Predicted	
		ω_1	ω_2
True	ω_1	1	1
	ω_2	0	8

Performance assessment: Multiclass & Supervised

Confusion Matrix

Accuracy

		Predicted				
		ω_1	ω_2	ω_3	$\dots$	ω_K
True	ω_1	T_{ω_1}	$F_{\omega_2 \rightarrow \omega_1}$	$F_{\omega_3 \rightarrow \omega_1}$	$\dots$	$F_{\omega_K \rightarrow \omega_1}$
	ω_2	$F_{\omega_1 \rightarrow \omega_2}$	T_{ω_2}	$F_{\omega_3 \rightarrow \omega_2}$	$\dots$	$F_{\omega_K \rightarrow \omega_2}$
	ω_3	$F_{\omega_1 \rightarrow \omega_3}$	$F_{\omega_2 \rightarrow \omega_3}$	T_{ω_3}	$\dots$	$F_{\omega_K \rightarrow \omega_3}$
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
	ω_K	$F_{\omega_1 \rightarrow \omega_K}$	$F_{\omega_2 \rightarrow \omega_K}$	$F_{\omega_3 \rightarrow \omega_K}$	$\dots$	T_{ω_K}

$$ACC = \frac{\sum_{i=1}^K T_{\omega_i}}{\sum_{i=1}^K T_{\omega_i} + \sum_{l=1}^K \sum_{c=1, l \neq c}^K F_{\omega_c \rightarrow \omega_l}} = \frac{\sum_{i=1}^K T_{\omega_i}}{N}$$

$$BACC = \frac{\sum_{i=1}^K \frac{T_{\omega_i}}{T_{\omega_i} + \sum_{c=1}^K F_{\omega_c \rightarrow \omega_i}}}{K}$$

$$BACC_W = \frac{\sum_{i=1}^K w_i \frac{T_{\omega_i}}{(T_{\omega_i} + \sum_{c=1}^K F_{\omega_c \rightarrow \omega_i})}}{W},$$

With w_i the weight for ω_i and $W = \sum w_i$

Performance assessment: Multiclass & Supervised

Confusion Matrix

		Predicted				
		ω_1	ω_2	ω_3	$\dots$	ω_K
True	ω_1	T_{ω_1}	$F_{\omega_2 \rightarrow \omega_1}$	$F_{\omega_3 \rightarrow \omega_1}$	$\dots$	$F_{\omega_K \rightarrow \omega_1}$
	ω_2	$F_{\omega_1 \rightarrow \omega_2}$	T_{ω_2}	$F_{\omega_3 \rightarrow \omega_2}$	$\dots$	$F_{\omega_K \rightarrow \omega_2}$
	ω_3	$F_{\omega_1 \rightarrow \omega_3}$	$F_{\omega_2 \rightarrow \omega_3}$	T_{ω_3}	$\dots$	$F_{\omega_K \rightarrow \omega_3}$
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
	ω_K	$F_{\omega_1 \rightarrow \omega_K}$	$F_{\omega_2 \rightarrow \omega_K}$	$F_{\omega_3 \rightarrow \omega_K}$	$\dots$	T_{ω_K}

Macro Measures

$$PR_i = \frac{T_{\omega_i}}{T_{\omega_i} + \sum_{l=1, l \neq i}^K F_{\omega_i \rightarrow \omega_l}}$$

$$SS_i = \frac{T_{\omega_i}}{T_{\omega_i} + \sum_{c=1, c \neq i}^K F_{\omega_c \rightarrow \omega_i}}$$

$$MSS = Macro\ SS = \frac{\sum_{i=1}^K SS_i}{K}$$

$$MPR = Macro\ PR = \frac{\sum_{i=1}^K PR_i}{K}$$

$$MF1score = 2 \frac{MPR \times MSS}{MSS + MPR}$$

Performance assessment: Multiclass & Supervised

Confusion Matrix

Micro Measures

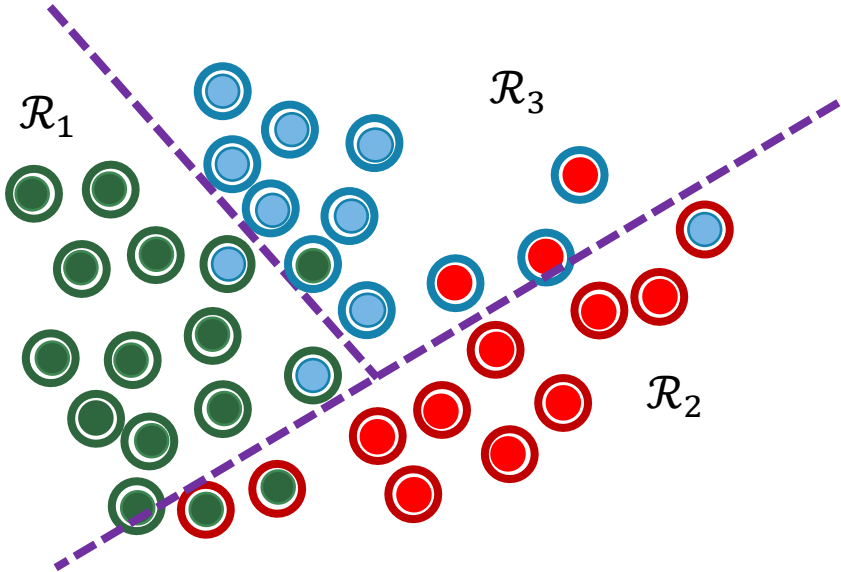
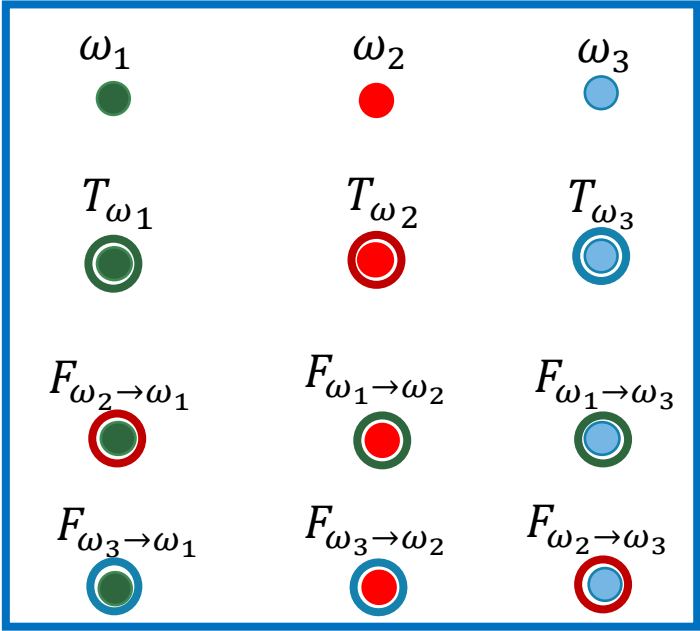
		Predicted				
		ω_1	ω_2	ω_3	$\dots$	ω_K
True	ω_1	T_{ω_1}	$F_{\omega_2 \rightarrow \omega_1}$	$F_{\omega_3 \rightarrow \omega_1}$	$\dots$	$F_{\omega_K \rightarrow \omega_1}$
	ω_2	T_{ω_1}	T_{ω_2}	$F_{\omega_3 \rightarrow \omega_2}$	$\dots$	$F_{\omega_K \rightarrow \omega_2}$
	ω_3	$F_{\omega_1 \rightarrow \omega_3}$	$F_{\omega_2 \rightarrow \omega_3}$	T_{ω_3}	$\dots$	$F_{\omega_K \rightarrow \omega_3}$
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
	ω_K	$F_{\omega_1 \rightarrow \omega_K}$	$F_{\omega_2 \rightarrow \omega_K}$	$F_{\omega_3 \rightarrow \omega_K}$	$\dots$	T_{ω_K}

$$mSS = \text{Micro } SS = \frac{\sum_{i=1}^K T_{\omega_i}}{N}$$

$$mPR = \text{Micro } PR = \frac{\sum_{i=1}^K T_{\omega_i}}{N}$$

$$mF1score = \frac{\sum_{i=1}^K T_{\omega_i}}{N} = ACC$$

Multiclass Example



		Predicted		
		ω_1	ω_2	ω_3
True	ω_1	$T_{\omega_1} = 11$	$F_{\omega_2 \rightarrow \omega_1} = 2$	$F_{\omega_3 \rightarrow \omega_1} = 1$
	ω_2	$F_{\omega_1 \rightarrow \omega_2} = 0$	$T_{\omega_2} = 8$	$F_{\omega_3 \rightarrow \omega_2} = 3$
	ω_3	$F_{\omega_1 \rightarrow \omega_3} = 2$	$F_{\omega_2 \rightarrow \omega_3} = 1$	$T_{\omega_3} = 7$

Multiclass Example

		Predicted		
		ω_1	ω_2	ω_3
True	ω_1	$T_{\omega_1} = 11$	$F_{\omega_2 \rightarrow \omega_1} = 2$	$F_{\omega_3 \rightarrow \omega_1} = 1$
	ω_2	$F_{\omega_1 \rightarrow \omega_2} = 0$	$T_{\omega_2} = 8$	$F_{\omega_3 \rightarrow \omega_2} = 3$
	ω_3	$F_{\omega_1 \rightarrow \omega_3} = 2$	$F_{\omega_2 \rightarrow \omega_3} = 1$	$T_{\omega_3} = 7$

$N = 35, K = 3$

$$ACC = \frac{\sum_{i=1}^K T_{\omega_i}}{\sum_{i=1}^K T_{\omega_i} + \sum_{l=1}^K \sum_{c=1, l \neq c}^K F_{\omega_c \rightarrow \omega_l}} = \frac{\sum_{i=1}^K T_{\omega_i}}{N}$$
$$= \frac{26}{35} \approx 0.74 \rightarrow \approx 74\%$$

$$BACC = \frac{\sum_{i=1}^K \frac{T_{\omega_i}}{T_{\omega_i} + \sum_{c=1}^K F_{\omega_c \rightarrow \omega_i}}}{K} = \frac{\frac{11}{14} + \frac{8}{11} + \frac{7}{10}}{3}$$
$$\approx 0.74 \rightarrow \approx 74\%$$

$$BACC_W = \frac{\sum_{i=1}^K w_i \frac{T_{\omega_i}}{(T_{\omega_i} + \sum_{c=1}^K F_{\omega_c \rightarrow \omega_i})}}{W} = \frac{0.40 \frac{11}{14} + 0.31 \frac{8}{11} + 0.28 \frac{7}{10}}{1} \approx 0.74 \rightarrow 74\%,$$

being $w_1 = \frac{14}{35} = 0.40, w_2 = \frac{11}{35} \approx 0.31, w_3 = \frac{10}{35} \approx 0.28$ the relative freq. of $\omega_1, \omega_2, \omega_3$ respectively, and $W \approx 1$.

Multiclass Example

		Predicted		
		ω_1	ω_2	ω_3
True	ω_1	$T_{\omega_1} = 11$	$F_{\omega_2 \rightarrow \omega_1} = 2$	$F_{\omega_3 \rightarrow \omega_1} = 1$
	ω_2	$F_{\omega_1 \rightarrow \omega_2} = 0$	$T_{\omega_2} = 8$	$F_{\omega_3 \rightarrow \omega_2} = 3$
	ω_3	$F_{\omega_1 \rightarrow \omega_3} = 2$	$F_{\omega_2 \rightarrow \omega_3} = 1$	$T_{\omega_3} = 7$

$K = 3$

Macro Measures

$$MSS = \frac{\sum_{i=1}^K SS_i}{K} = \frac{0.78 + 0.73 + 0.70}{3} \approx 0.74 \rightarrow 74\%$$

$$MPR = \frac{\sum_{i=1}^K PR_i}{K} = \frac{0.85 + 0.73 + 0.62}{3} \approx 0.73 \rightarrow 73\%$$

$$MF1score = 2 \frac{MPR \times MSS}{MSS + MPR} \approx 0.73 \rightarrow 73\%$$

$$PR_1 = \frac{T_{\omega_1}}{T_{\omega_1} + \sum_{l=1, l \neq 1}^3 F_{\omega_1 \rightarrow \omega_l}} = \frac{11}{11 + (0 + 2)} = \frac{11}{13} \approx 0.85$$
$$PR_2 = \frac{T_{\omega_2}}{T_{\omega_2} + \sum_{l=1, l \neq 2}^3 F_{\omega_2 \rightarrow \omega_l}} = \frac{8}{8 + (2 + 1)} = \frac{8}{11} \approx 0.73$$
$$PR_3 = \frac{T_{\omega_3}}{T_{\omega_3} + \sum_{l=1, l \neq 3}^3 F_{\omega_3 \rightarrow \omega_l}} = \frac{7}{7 + (1 + 3)} = \frac{7}{11} \approx 0.62$$

$$SS_1 = \frac{T_{\omega_1}}{T_{\omega_1} + \sum_{c=1, c \neq 1}^3 F_{\omega_c \rightarrow \omega_1}} = \frac{11}{11 + (2 + 1)} = \frac{11}{14} \approx 0.78$$
$$SS_2 = \frac{T_{\omega_2}}{T_{\omega_2} + \sum_{c=1, c \neq 2}^3 F_{\omega_c \rightarrow \omega_2}} = \frac{8}{8 + (0 + 3)} = \frac{8}{11} \approx 0.73$$
$$SS_3 = \frac{T_{\omega_3}}{T_{\omega_3} + \sum_{c=1, c \neq 3}^3 F_{\omega_c \rightarrow \omega_3}} = \frac{7}{7 + (2 + 1)} = \frac{7}{10} = 0.70$$

Performance assessment: Resume

- Macro
 - Average measures of the performance for each class
 - Class level measures → Each class is equally important, regardless of the class size → **Blind to imbalance**
- Micro
 - The same as accuracy → **Also blind to class imbalance**
- Classe imbalance
 - Multiclass → **Use $BACC_W$ or $BACC$**
 - Binary → **Use $F1score$, $BACC_W$ or $BACC$**
- **Other performance measures exist...** (see for example <https://arxiv.org/pdf/2008.05756>)

Classifier Evaluation: Data partitioning

- Determination of reliable **estimates of the classifier performance is an essential task** to assess its usefulness and compare it with alternative solutions
- The performance should, when possible, be **evaluated in an independent testing/validation set**

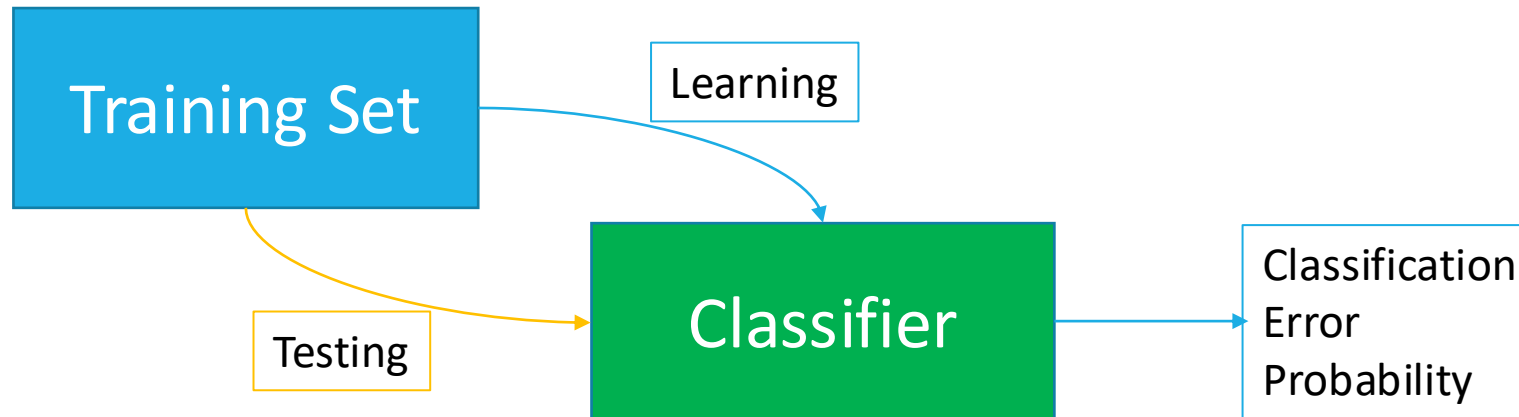
Data partitioning

How to create training & testing sets?

- **Resubstitution Method**
- **Holdout Method**
- **K-fold Cross Validation**
- **Bootstrap Method**

Data Partitioning: Resubstitution

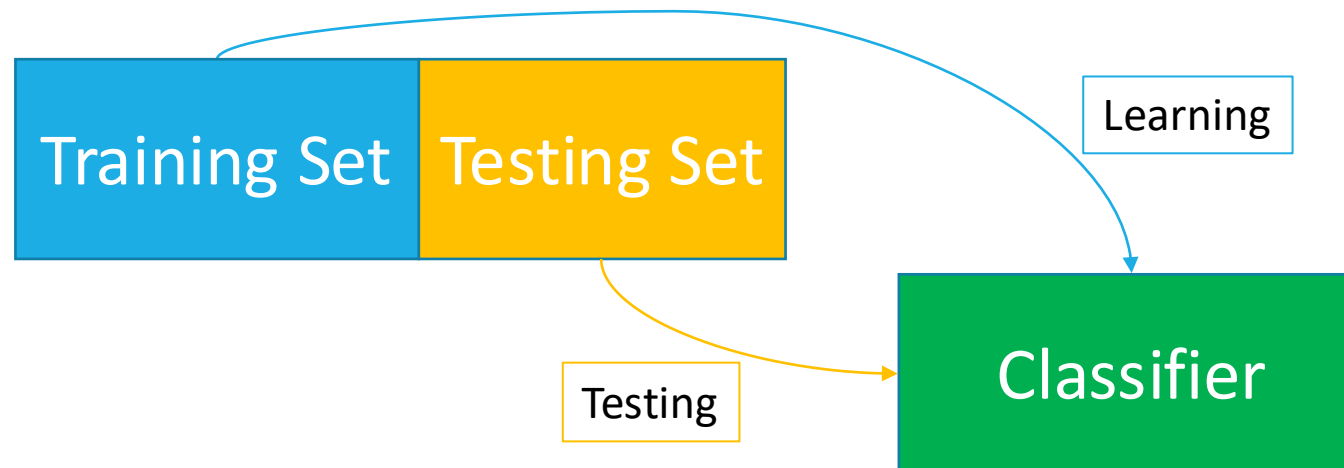
- The **whole dataset is used for training and testing**
- Corresponds to setting the **error estimate on the training set** (apparent error)
 - The estimates of error are thus **optimistic and have low variance**
 - The effect is more **noticeable as a smaller training set is** (overfitting)



Classifier Evaluation: Holdout

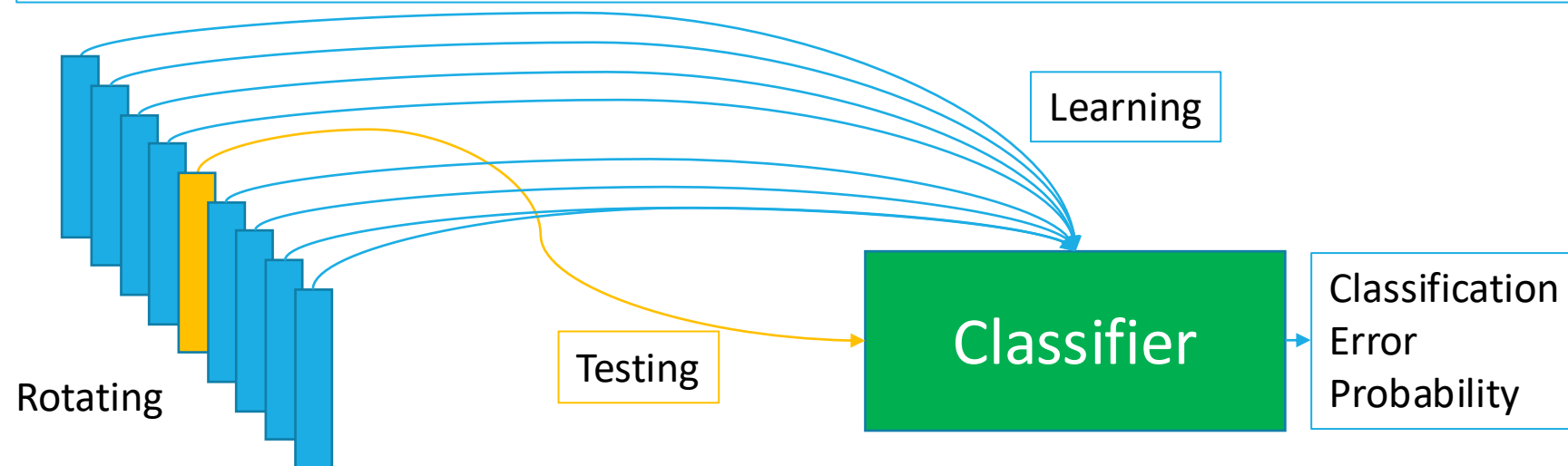
- The total available n samples are (randomly or not) **divided into two disjoint sets**. E.g.:
 - 50% training
 - 50% testing
- More common:
 - 70% training
 - 30% testing
- The error estimate is obtained from the testing set
 - **Improve estimation by doing several runs** with different data for training & testing.

Sequences may be important (e.g., time series)



Classifier Evaluation: K-Fold/Leave-one-out

- Divide data into $k > 1$ subsets of randomly selected patterns
- Keeps one part (fold) with n/k patterns for testing and the remaining $(n - 1)$ parts for training → This is performed in a rotating way k times.
 - ❖ Average/median testing performance (e.g. average sensitivity: $\widehat{SS}_{e_t} = \frac{1}{k} \sum_{i=1}^k \widehat{SS}_{e_{ti}}$, being $\widehat{SS}_{e_{ti}}$ the sensitivity for the i -th testing fold)
 - ❖ Variability of the testing performance (e.g., variance of the specificity $\sigma_{SP_{et}}^2 = \frac{1}{k-1} \sum_{i=1}^k (\widehat{SP}_{e_{ti}} - \widehat{SP}_{e_t})^2 \rightarrow$ Standard deviation: $\sigma_{SP_{et}} = \sqrt{\sigma_{SP_{et}}^2}$)
 - ❖ Report always Average/median $\pm$ Variability (e.g., $\widehat{SS}_{e_t} \pm \sigma_{SS_{et}}$)



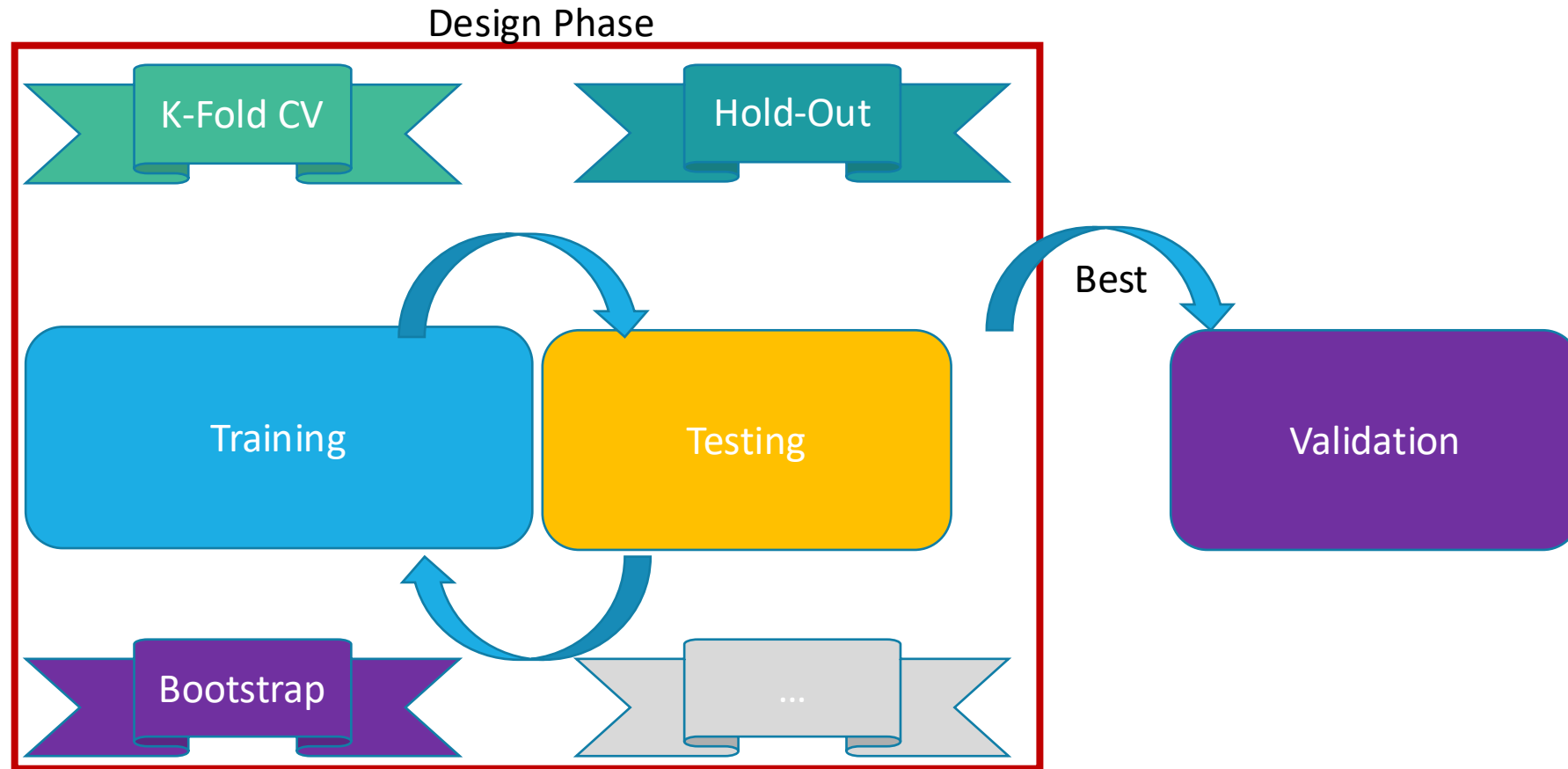
- $k=2 \rightarrow$ Similar to Hold Out 50-50
- $K=n \rightarrow$ Leave-one-out (Low bias Very high variance)

Classifier Evaluation: Bootstrap

- Generation of data portions by randomly selecting existing samples with equal probability
- The same sample can appear several times in training → **Sampling with Replacement**



Three Set Validation



- Real World Evaluation
- Discard the influence originated by selection and multi-testing

Classifier Evaluation

Design data

Validation data

Hold-Out

Training

Testing

K-Fold CV

Training

Testing

Bootstrap

Testing

Training

- Never Used Data
- Future Data

Bias & Variance

- Training set is finite and a subset of all possible cases.
- Different training sets → Different classifier parameters → Different performance → Compute average performance & variance
- Total error rate of a classifier can be decomposed into two parts:
 - Error rate due to the average performance of the classifier → **Bias**
 - Error rate due to training set changes → **Variance**

Bias & Variance

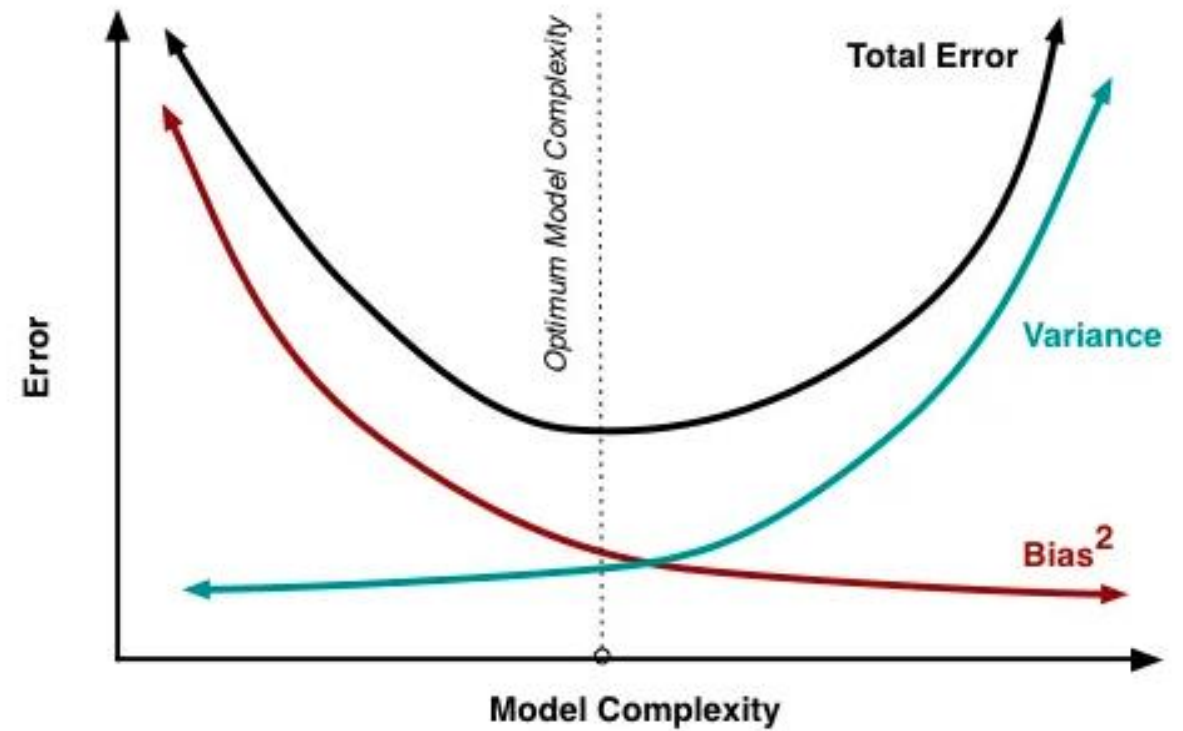
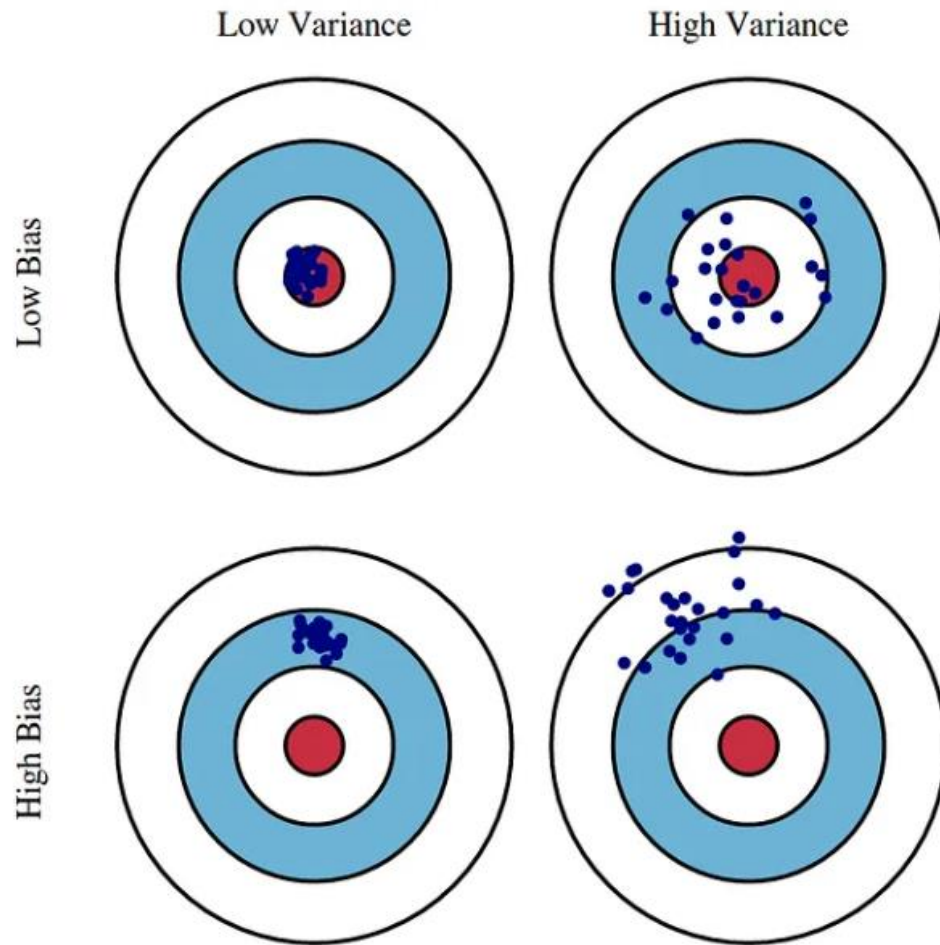
- **Bias**

- How far is the average performance from optimal?

- **Variance**

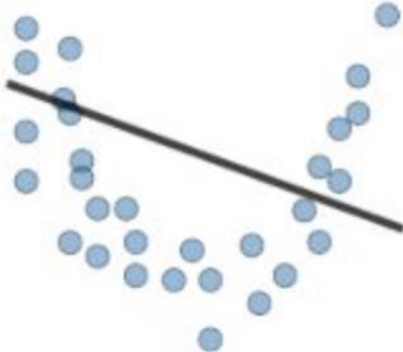
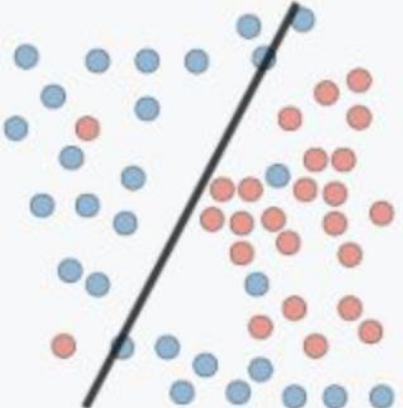
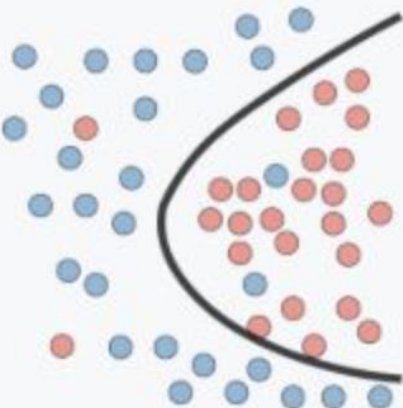
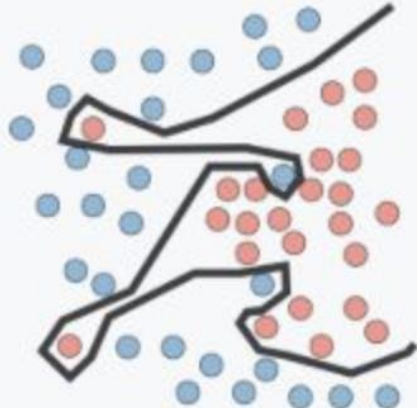
- How much does the performance vary over different training sets? or
- How much does the decision boundary vary over different training sets?

Bias & Variance



<https://towardsdatascience.com/random-forests-and-the-bias-variance-tradeoff-3b77fee339b4>

Error Bias & Variance

	Underfitting	Just right	Overfitting
Symptoms	• High training error • Training error close to test error • High bias	• Training error slightly lower than test error	• Very low training error • Training error much lower than test error • High variance
Regression illustration			
Classification illustration			

Summary

- ❖ Linear and generalized decision functions
- ❖ Multiclass problems are addressed through combinations of binary classification tasks
 - ❖ One-vs-All (OAA)
 - ❖ One-vs-One (OAO)
- ❖ Performance evaluation
 - ❖ Binary
 - ❖ Multiclass
- ❖ Macro versus Micro measures
- ❖ Data partitioning strategies
- ❖ Bias-Variance Dilemma